DEPI · Digital Egypt Pioneers Initiative

EDUVANCE AI

Project Planning & Management

An Agentic AI Platform for Transforming Training Materials into Personalized Learning Experiences

PROGRAM	Digital Egypt Pioneers Initiative
TRACK	AI & Data Science — Agentic AI and Generative AI System Developer
CONTENT	Proposal · Plan · Roles · Risk Management · KPIs

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Project Proposal

Overview, objectives, scope

Project Definition and Academic Objectives

Problem statement

Training materials are often distributed across lengthy documents, slides, and unstructured notes. Learners must organize these resources, identify prerequisite knowledge, and find ways to assess their understanding.

Eduvance AI addresses this problem by transforming uploaded materials into a structured learning experience that combines explanations, videos, source-grounded tutoring, assessments, and progress-based recommendations.

Main objective

Design, implement, and evaluate an integrated AI-assisted learning platform that converts a bounded collection of training documents into a personalized learning experience.

Specific objectives

1. Extract and organize concepts from uploaded documents.
2. Generate a course structure with modules, lessons, prerequisites, and learning objectives.
3. Produce short educational videos using generated scripts, slides, narration, and captions.
4. Provide a tutor that answers questions using the uploaded materials and cites its sources.
5. Generate assessments and provide appropriate feedback.
6. Track completion and assessment performance to recommend revision activities.
7. Suggest a practical project aligned with the learner's completed topics.

Academic contribution

The project should demonstrate:

- An integrated workflow connecting content understanding, generation, tutoring, and assessment.
- Traceability from generated learning materials back to their sources.
- Explainable personalization based on observable learner activity.
- An evaluation of output quality, reliability, usability, and operating cost.

The team should distinguish its own contributions—system design, orchestration, interfaces, evaluation, and personalization rules—from capabilities supplied by external models. It should not claim to have trained a new foundation model or proven improved learning outcomes without suitable evidence.

2. Scope and Priorities

A DEPI capstone project needs a complete, demonstrable workflow within a manageable scope.

Capability	Required prototype scope	Deferred enhancement
Document upload	Text-based PDF files; initial limit of 3 files and 50 pages per course	Scanned PDFs, OCR, slides, and additional formats
Content understanding	Extract concepts, estimate difficulty, and identify prerequisites	Advanced knowledge graphs
Course design	Generate approximately 3 modules and 6 lessons, with editable objectives	Large courses and complex curriculum optimization

Capability	Required prototype scope	Deferred enhancement
Video generation	Short slide-based videos with narration and captions	Avatars and advanced animation
AI tutor	Course-specific RAG chat with source references	Voice conversations and external web research
Assessment	MCQ and true/false questions with automatic scoring; practical exercises with rubric-based feedback	Automatic execution and grading of submitted code
Progress tracking	Completion, quiz results, concept-level performance indicators	Predictive learner modeling
Personalization	Transparent rules recommending lessons to review	Reinforcement learning or advanced adaptive sequencing
Project mentor	Project brief, implementation steps, and assessment rubric	Autonomous project supervision
Deployment	Hosted demonstration with basic accounts and user isolation	Enterprise-scale infrastructure and integrations

Language assumption: Validate one primary content and interface language first. Select it in Week 1; treat additional language support as a stretch objective.

Priority order

- **P0 — Foundation:** Upload, extraction, course structure, data storage, and a basic interface.
- **P1 — Required learning experience:** Tutor, simple videos, assessments, progress, and a basic project mentor.
- **P2 — Optional improvements:** Additional languages, richer visuals, broader file support, and advanced personalization.

P0 and P1 form the capstone baseline. P2 work begins only after the baseline passes integration testing.

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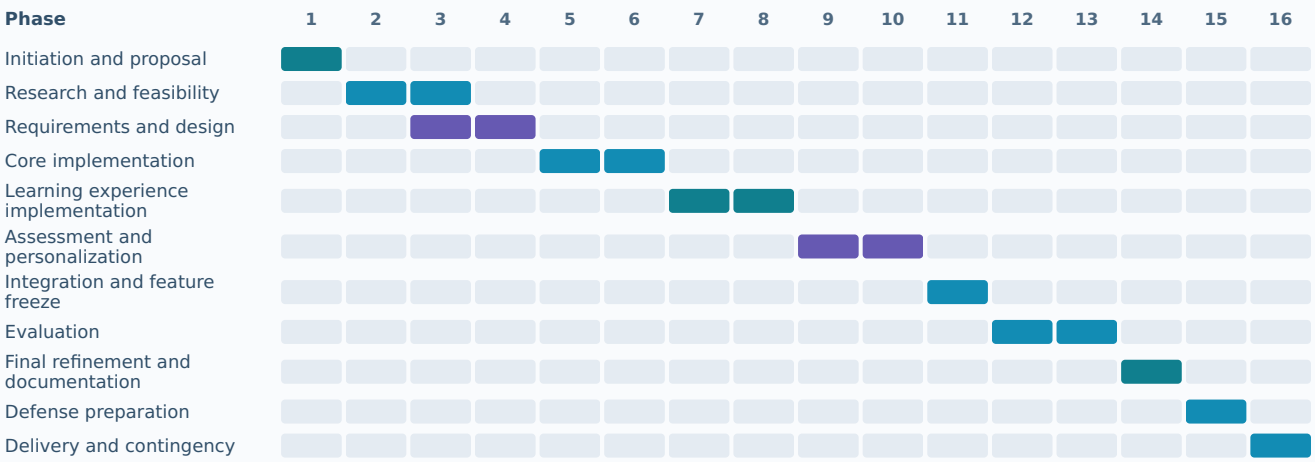
Project Plan

Lifecycle, Gantt, milestones, deliverables, resource allocation

End-to-End Project Lifecycle

PROJECT LIFECYCLE — GANTT VIEW

Relative week numbers reproduced from the original Project Plan.



Phase	Weeks	Main activities	Deliverables and exit criteria
1. Initiation and proposal	1	Confirm academic requirements, define the problem, assign roles, agree on scope	Approved proposal, project charter, initial backlog, and supervisor expectations
2. Research and feasibility	2–3	Review related work; test extraction, retrieval, and video generation; select implementation approach	Literature review draft, feasibility results, initial evaluation dataset, and architecture decision
3. Requirements and design	3–4	Define use cases, data structures, interfaces, agent contracts, and acceptance criteria	Requirements specification, architecture diagram, database design, and interface prototype
4. Core implementation	5–6	Build upload, extraction, source mapping, course generation, and course review	Documents can be converted into an editable, source-linked course outline
5. Learning experience implementation	7–8	Integrate tutor, lesson display, and short video generation	One complete lesson works from upload through video playback and tutoring
6. Assessment and personalization	9–10	Implement quizzes, scoring, progress rules, and project recommendations	All required capabilities work individually and are ready for final integration
7. Integration and feature freeze	11	Connect the complete workflow; handle failures; prepare demonstration deployment	End-to-end prototype with required features frozen
8. Evaluation	12–13	Run quality benchmarks, user testing, baseline comparison, and reliability tests	Evaluation results, defect log, and documented limitations
9. Final refinement and documentation	14	Fix critical defects, finalize report, prepare setup and user guides	Stable release candidate and complete report draft

Phase	Weeks	Main activities	Deliverables and exit criteria
10. Final presentation preparation	15	Prepare slides, record backup demo, rehearse presentation and questions	Submission package and rehearsed presentation
11. Delivery and contingency	16	Submit, demonstrate, hand over artifacts, and resolve remaining blockers	Final submission, working demonstration, and archived project materials

Literature review and report writing continue throughout development. Week 16 is primarily contingency and delivery time, not a planned feature-development period.

7. Milestones and Deliverables

Milestone	Deadline	Required evidence
M1 — Proposal approved	End of W1	Clear problem, objectives, scope, roles, and academic expectations
M2 — Feasibility confirmed	End of W3	Successful extraction, retrieval, and short-video experiments; initial cost estimates
M3 — Design baseline complete	End of W4	Architecture, database model, component contracts, interface prototype
M4 — Content-to-course pipeline	End of W6	Uploaded documents produce an editable course with source references
M5 — First interactive lesson	End of W8	Lesson video and tutor work within the application
M6 — Required capabilities implemented	End of W10	Assessments, progress, and project mentoring connected to the course
M7 — Feature freeze	End of W11	Full learner journey works on the demonstration environment
M8 — Evaluation complete	End of W13	Metrics, baseline comparison, user-test findings, and limitations
M9 — Report and prototype ready	End of W14	Reviewed report draft and stable release candidate
M10 — Presentation ready	End of W15	Final slides, backup recording, rehearsal, and anticipated questions
M11 — Final delivery	W16	Submitted report, code, demo, documentation, and required academic forms

Critical path

Approved scope → extraction and source mapping → course structure → lesson outputs → integrated learner journey → evaluation → final report and presentation.

Interface development and video feasibility experiments can proceed in parallel once their input and output formats are agreed.

Workload estimate

Total available capacity:

4 students × 15 hours/week × 16 weeks = 960 hours

Member	Planned hours	Reserved capacity
Member 1	185	55
Member 2	195	45
Member 3	195	45
Member 4	193	47
Total	**768**	**192**

The **20% reserve** covers integration problems, coursework conflicts, rework, and final presentation preparation. Planned hours include research, meetings, testing, and documentation.

10. Project Control and Communication

- **Weekly planning:** Assign tasks according to remaining capacity and dependencies.
- **Weekly supervisor update:** Present completed work, decisions needed, risks, and next steps.
- **Fortnightly demonstration:** Show working functionality.
- **Single task board:** Track owner, priority, dependency, due date, status, and acceptance evidence.
- **Shared decision log:** Record scope changes, architecture choices, and reasons.
- **Feature freeze:** No new required features after Week 11 unless necessary to meet an existing acceptance criterion.

A task is complete when it works, passes appropriate checks, has been reviewed, and is documented sufficiently for another team member to understand and maintain it.

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Task Assignment & Roles

Original WBS ownership, responsibilities and RACI

Work Breakdown Structure

Each package has one accountable owner. Contributors may review or support the work without duplicating ownership.

WBS	Work package and actionable tasks	Owner	Schedule	Main dependency
1.1	Confirm DEPI program requirements, deadlines, and marking criteria	Member 1	W1	None
1.2	Write problem statement, objectives, scope, and proposal	Member 1	W1	Academic requirements
1.3	Review related systems and research; build comparison table	Member 2	W2–3	Defined research scope
1.4	Define evaluation questions, datasets, and metrics	Member 2	W2–3	Project objectives
2.1	Design architecture, data model, and integration contracts	Member 1	W3–4	Feasibility findings
2.2	Design user journeys, wireframes, and reusable interface components	Member 4	W3–4	Requirements
2.3	Test video narration, captions, rendering time, and cost	Member 3	W2–3	Representative lesson
3.1	Set up repository, environments, database, accounts, and storage	Member 1	W3–5	Architecture
3.2	Implement upload validation, extraction, and source references	Member 2	W4–5	Sample documents
3.3	Implement concept extraction and course generation	Member 2	W5–6	Extracted content
3.4	Build upload, processing-status, and course-review screens	Member 4	W5–6	Interface and API contracts
4.1	Implement workflow execution, status tracking, retries, and saved outputs	Member 1	W5–8	Component contracts
4.2	Implement retrieval, grounded answers, citations, and abstention	Member 2	W7–8	Source-linked content
4.3	Implement script, slide, narration, caption, and video pipeline	Member 3	W5–8	Approved lesson structure
4.4	Build lesson player, tutor chat, and source display	Member 4	W7–8	Tutor and video interfaces
5.1	Implement question generation, answer keys, and exercise rubrics	Member 3	W9	Lesson objectives
5.2	Implement assessment storage, scoring, and learning-event APIs	Member 1	W9–10	Assessment schema
5.3	Implement progress calculations and revision rules	Member 1	W10	Learning events
5.4	Implement bounded project recommendations and project rubrics	Member 3	W10	Course concepts
5.5	Build quiz, results, progress, and project screens	Member 4	W9–10	Relevant APIs
6.1	Integrate the full workflow and coordinate end-to-end tests	Member 4	W11	Required components
6.2	Run tutor and course-quality evaluation and baseline comparison	Member 2	W12–13	Frozen prototype and dataset
6.3	Evaluate video, assessment, and project-output quality	Member 3	W12–13	Generated outputs
6.4	Run usability sessions and summarize task-completion results	Member 4	W12–13	Test participants
6.5	Deploy demo; test access isolation, recovery, and operating limits	Member 1	W11–14	Integrated prototype
7.1	Consolidate report, references, results, and limitations	Member 1	W2–14	Contributions from all members
7.2	Prepare presentation structure and demonstration materials	Member 4	W14–15	Stable prototype

WBS	Work package and actionable tasks	Owner	Schedule	Main dependency
7.3	Verify submission package and complete handover	Member 1	W16	Final acceptance

Every work package should be divided into smaller tasks with an expected output and acceptance criterion before entering a weekly work plan.

6. Roles and Responsibility Matrix

Member	Role	Primary ownership	Academic responsibility
Member 1	Agentic AI Orchestration & Integration Lead	Planning, integration contracts, accounts, storage, workflow controller, scoring, progress logic, deployment	Proposal, architecture and implementation sections, report consolidation
Member 2	RAG & Knowledge Intelligence Engineer	Extraction, concept analysis, course design, retrieval, tutor, grounding evaluation	Related work, content/RAG methodology, experimental design
Member 3	Generative AI Content & Assessment Engineer	Video pipeline, question generation, exercise rubrics, project mentor	Media and assessment methodology, output-quality results
Member 4	AI Evaluation & Personalization Engineer	Interface, learner journey, dashboards, integration-test coordination, usability testing	Interface design, usability results, demonstration materials

Ownership rules

- Each member implements and tests their own components.
- Member 4 coordinates integration testing; they do not carry the entire team's testing workload.
- Member 1 consolidates the report; every member writes and verifies their own technical sections.
- Shared interfaces are agreed before implementation.
- A different member reviews each major component before integration.

RACI matrix

A = Accountable, R = Responsible, C = Consulted, I = Informed.

Activity	Member 1	Member 2	Member 3	Member 4
Project planning and scope	A/R	C	C	C
Backend and deployment	A/R	C	C	C
Content, course generation, and tutor	C	A/R	C	I
Video, assessments, and project mentor	C	C	A/R	I
Progress calculations	A/R	C	C	C
Interface and learner experience	C	C	C	A/R
Integration-test coordination	C	C	C	A/R
AI evaluation methodology	C	A/R	C	C
Final report consolidation	A/R	R	R	R
Presentation preparation	C	C	C	A/R
Submission coordination	A/R	C	C	C

Supervisor approval is an external academic gate and does not replace internal ownership.

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Risk Assessment & Mitigation Plan

Risks, impacts, mitigations and owners

Risk Management

Risk	Impact	Mitigation	Owner
Scope exceeds available student time	High	Freeze baseline early; defer optional features	Member 1
Video generation is slow or expensive	High	Use short slide-based videos; measure cost early; reuse saved outputs	Member 3
Tutor produces unsupported answers	High	Require sources, evaluate grounding, and implement abstention	Member 2
Document extraction fails	High	Restrict supported inputs and validate representative files early	Member 2
Components work separately but fail together	High	Agree contracts early and integrate a complete lesson by W8	Member 1
Exams or coursework reduce availability	High	Track capacity weekly and preserve contingency	Member 1
External AI services become unavailable	Medium–High	Save generated outputs, handle failure clearly, and prepare a recorded demo	Member 1
Report writing is postponed	High	Write sections during each phase and review weekly	Member 1
Evaluation participants are unavailable	Medium	Recruit by W8 and schedule sessions before W12	Member 4
Sensitive or unauthorized documents are uploaded	High	Use permitted materials and enforce account isolation	Member 1

For user testing, explain what data will be collected, minimize personal information, and follow the DEPI program's applicable approval process.

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Key Performance Indicators (KPIs)

Existing evaluation targets and acceptance criteria

Academic Evaluation and Acceptance Criteria

Evaluation should answer specific questions, rather than rely only on a successful demonstration.

Evaluation question	Method	Initial acceptance target
Does the generated course reflect the source material?	Review the six demonstration lessons against source passages and learning objectives	Every lesson has objectives and traceable sources; no unresolved major factual error
Does the tutor provide grounded answers?	Fixed set of 40 in-scope questions, reviewed against reference answers	At least 90% correct and source-supported responses
Does the tutor recognize missing information?	20 questions whose answers are absent from the uploaded material	Appropriate abstention or clarification on at least 18 questions
Does retrieval improve grounding?	Compare the same model with and without retrieved course context, using the same question set	Report measured differences in correctness and unsupported claims
Are generated assessments usable?	Review at least 30 questions for correctness, ambiguity, and objective alignment	At least 90% pass the review rubric; fix rejected questions before demonstration
Are videos understandable and accurate?	Review narration, captions, readability, and alignment across the six lessons	All demonstration videos pass the agreed quality checklist
Can users complete the main workflow?	Task-based test with approximately 8–12 participants	At least 80% complete the core tasks without direct assistance
Is personalization explainable?	Test predefined learner-result scenarios	Recommendations follow documented rules and identify the triggering result
Is the workflow dependable?	Repeat at least 20 representative end-to-end runs	At least 18 finish successfully within the defined retry policy
Is the prototype practical to operate?	Measure generation time, tutor latency, and cost per lesson/course	Report actual results against limits agreed after feasibility testing

These are planning targets, not claimed results. Keep a separate final evaluation set where possible, and record model versions, prompts, parameters, and test dates.

Small usability studies can support conclusions about usability and perceived usefulness. They do not, by themselves, establish that the platform improves learning outcomes.